

Disaster Recovery and the Term-Structure of Dividend Strips

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Motivation

- Recent empirical evidence shows that the term-structures of equity volatility and premia are downward sloping.
van Binsbergen Brandt Koijen (AER 2012), van Binsbergen Hueskes Koijen Vrugt (JFE 2013), van Binsbergen Koijen (2015)
- Also dividend risk is downward sloping.
Belo Collin-Dufresne Goldstein (JF 2014), Marfè (2014)
- Both these stylized facts usually fail to obtain in leading models.
- Additional testable implications to asset pricing frameworks can help us to understand the macroeconomic determinants of asset prices.

This Paper

- Rare disasters models “assume that there is a small probability of a large drop in consumption. [...] this large drop is extremely painful for investors and has a large impact on risk premia, and, through risk premia, on prices.” (Tsai Wachter 2015)
- Rare disasters models focus on the equity premium puzzle, the excess volatility puzzle, the return predictability, the option pricing.

Backus Chernov Martin (JF 2011), Barro (QJE 2006, AER 2009), Barro Ursúa (BPEA 2008), Berckman Jacobsen Lee (JFE 2011), Du (JFE 2011), Gabaix (AER 2012), Gourio (AER 2012), Julliard Ghosh (RFS 2012), Martin (ECMA 2013), Rietz (JME 1988), Wachter (JF 2013), ...

- Model assumptions focus on size and frequency of disasters.
- We argue that from a term-structure perspective post-disaster behavior is as much as important.

Gourio (AERpp 2008) and Nakamura Steinsson Barro Ursúa (AEJ:Macro 2013)

- **We show that post-disaster recovery affects the slope of dividend risk and, hence, the slope of equity.**

Intuition

- Consider the following dynamics

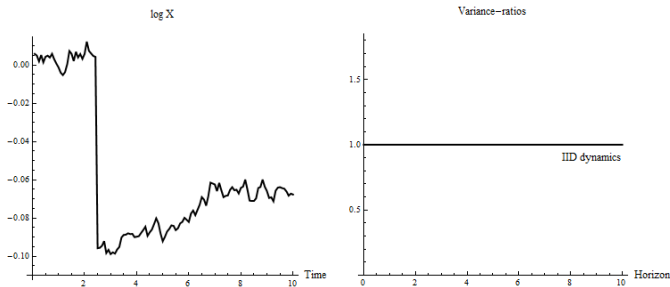
$$d \log X_t = \mu dt + \sigma dB_t + dJ_t$$

$$dJ_t = \xi_t dN_t$$

$$dN_t \sim \text{Poisson}(\lambda dt)$$

$$\xi_t \sim \text{NegExponential}(\eta)$$

- Sample path and term-structure of variance ratios:



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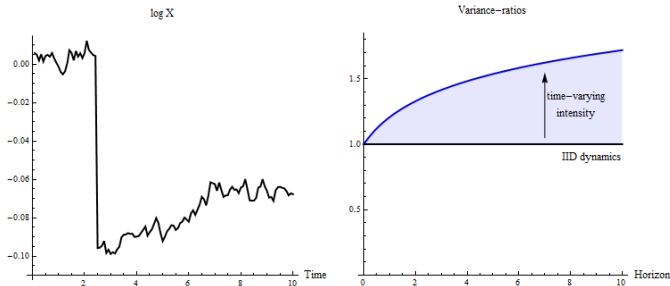
$$dJ_t = \xi_t dN_t$$

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$$d\lambda_t = \kappa(\bar{\lambda} - \lambda_t)dt + \rho\sqrt{\lambda_t}dW_t$$

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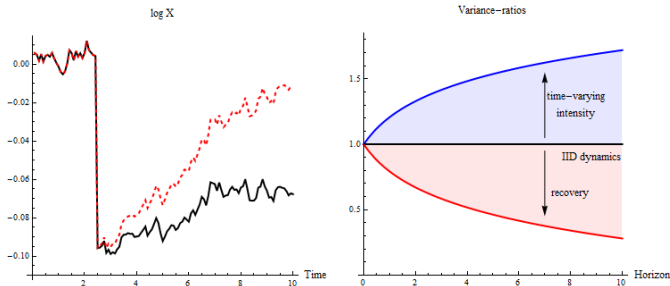
$$dJ_t = -\phi J_t dt + \xi_t dN_t$$

$$dN_t \sim \text{Poisson}(\lambda_t dt)$$

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- Sample path and term-structure of variance ratios:



Empirical Support

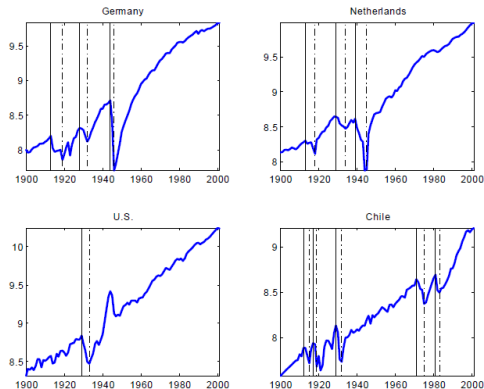
- We base our model on three stylized facts.

Empirical Support

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(1) Disasters are followed by recovery (about 50% in 5 years).

Gourio (AERpp 2008), Nakamura et al. (AEJ:Macro 2013)



Gourio (AER p&p 2008)

- International evidence of disaster recovery.

Empirical Support

- (2) Consumption and dividends are co-integrated.
Lettau Ludvigson (JFE 2005)

Empirical Support

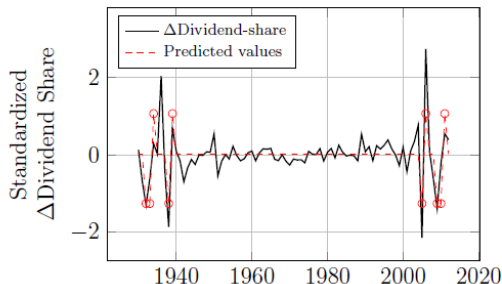
- (2) Consumption and dividends are co-integrated.

Lettau Ludvigson (JFE 2005)

- (3) Dividends load more on disasters and recover faster than consumption.

Longstaff Piazzesi (JFE 2004), Gourio (AER 2012)

- Dividend-share changes regressed on disaster and recovery dummies:
markers denote disaster and recovery events

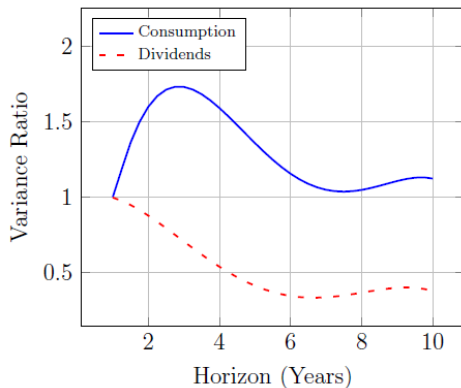


- Dividend-share moves negatively with disasters and positively with recoveries.
- Rationale for the term-structures of consumption and dividend risk.

Empirical Support

- These three stylized facts explain the gap between the slightly upward sloping consumption risk and the markedly downward sloping dividend risk.

Belo Collin-Dufresne Goldstein (JF 2014), Marfè (2014)



- Many models assume that long-horizon dividend risk is about one order of magnitude larger than in actual data.

Model and Main Results

- Endowment economy
- Representative agent with stochastic differential utility
affine dynamics + log-linearized return (Eraker Shaliastovich MF 2008)
- Joint dynamics of consumption and dividends:

$$\log C_t = x_t + z_{1t},$$

$$\log D_t = x_t + z_{2t} + \log d_0, \quad d_0 \in (0, 1),$$

with

$$dx_t = (\mu_t - \frac{1}{2}\sigma_x^2 + \omega z_{1t})dt + \sigma_x dW_{xt},$$

$$dz_{1t} = -\phi_z z_{1t}dt + \xi_t dN_{zt},$$

$$dz_{2t} = -\phi_z z_{2t}dt + \alpha \xi_t dN_{zt} + \sigma_z dW_{zt},$$

$$d\lambda_t = \phi_\lambda (\bar{\lambda} - \lambda_t)dt + \sigma_\lambda \sqrt{\lambda_t} dW_{\lambda t},$$

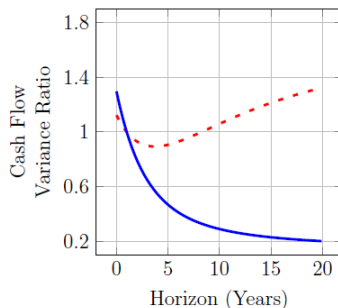
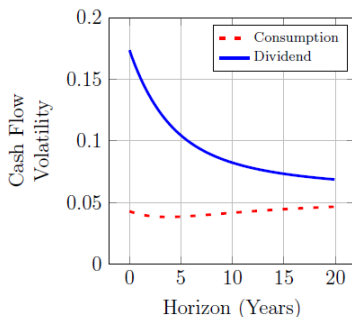
$$d\mu_t = \phi_\mu (\bar{\mu} - \mu_t)dt + \sigma_\mu dW_{\mu t}.$$

- Dividend strip price with maturity τ :

$$S_{t,\tau} = \mathbb{E}_t^{\mathbb{Q}}[e^{-\int_0^\tau r_{t+u}du} D_{t+\tau}]$$

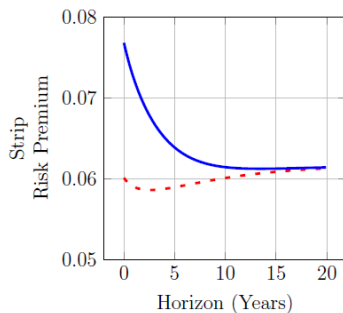
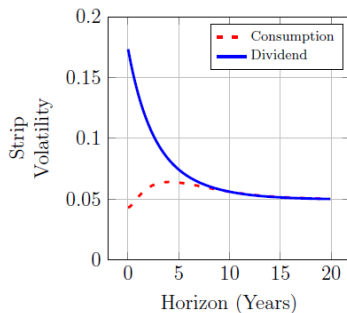
→ Itô's Lemma provides the term-structures of premia and volatility.

Model and Main Results



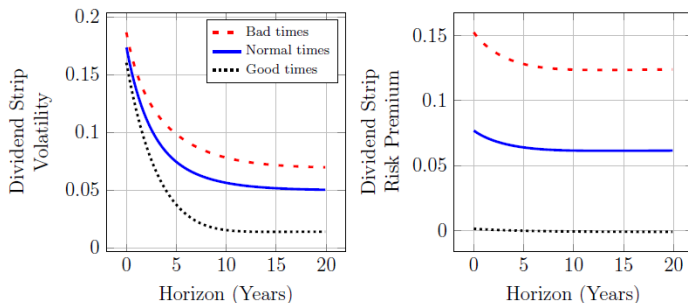
- Model calibration matches –among other stylized facts– the slightly upward sloping consumption risk and the markedly downward sloping dividend risk.

Model and Main Results



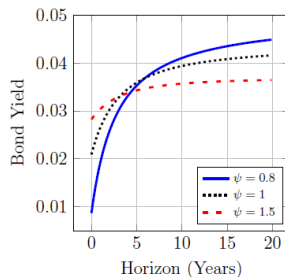
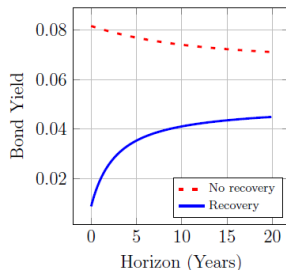
- The slopes of equity volatility and premia are negative (baseline calibration: $EIS = 0.8$, $RRA = 5.5$).

Model and Main Results



- Consistently with van Binsbergen et al. (JFE 2013), the slope of equity premia is pro-cyclical: approximatively flat in good times and markedly negative in bad times.

Model and Main Results

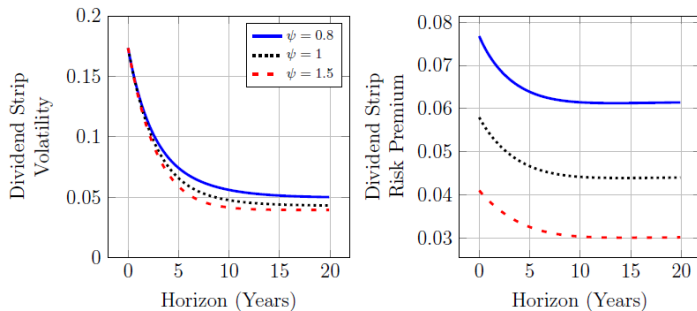


- Recovery leads to positive and increasing real yields, consistently with TIPs data. Slope is robust to EIS.

Conclusion

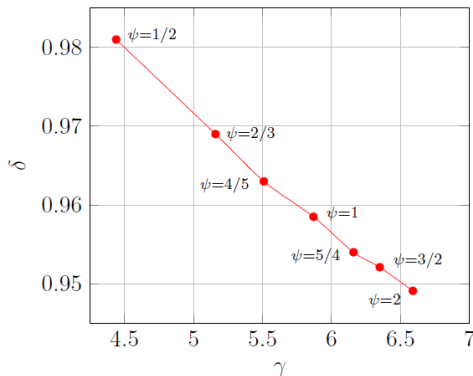
- Leading models overestimate long-horizon dividend risk.
- Disaster recovery can help to explain the term-structures of fundamentals' risk and, in turn, the term-structures of equity and interest rates.
- Model can reconcile a quantitative solution of the risk-free rate and equity premium puzzle with the term-structures of
 - consumption risk,
 - dividend risk,
 - equity volatility and premia,
 - real interest rates.
- Model results are robust to the specification of the elasticity of intertemporal substitution.

Other Model Predictions



- The negative slopes of equity volatility and premia are robust to the EIS specification.

Other Model Predictions



- Given downward sloping term-structures, model can fit the risk-free rate (0.6%) and the equity premium (6.5%) for any $EIS \in (1/2, 2)$ and low RRA.

Other Model Predictions

- Long-horizon predictability of excess returns by the price-dividend ratio:

Panel A – Model simulations with realized disasters					
Horizon (years)	1	2	3	5	7
coefficient	-0.287	-0.479	-0.604	-0.715	-0.728
t-stat	-16.1	-14.5	-12.6	-9.3	-6.9
adj-R ²	0.27	0.24	0.20	0.14	0.11

Panel B – Model simulations without realized disasters					
Horizon (years)	1	2	3	5	7
coefficient	-0.310	-0.519	-0.657	-0.789	-0.794
t-stat	-19.4	-16.7	-14.2	-10.5	-7.8
adj-R ²	0.34	0.28	0.24	0.17	0.12

- Negative and significant intertemporal relationship. Explanatory power decreases with the horizon.

Model Extension

- Nakamura et al. (AEJ:Macro 2013) document that consumption disasters unfold for a few years.
- Slightly modified dynamics: recovery still leads to downward sloping term-structures.

